

Gauthier Bouyjou

2D/3D Computer Vision/Graphics Software Engineer
RQTH

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📁 Portfolio : <https://hiergaut.github.io/> 📍 Location : France ☆ <https://github.com/hiergaut> 📅 Birth : 1993
🇫🇷 Nationality : French



SUMMARY

C++/Python software engineer specialized in **real-time systems and image processing**, with a strong passion for **3D software** and performance-critical applications for cinema, simulation, medical, aerospace and defense domains. Committed to **high-quality software development**.

WORK EXPERIENCE

Computer Graphics, Image Processing Software Engineer, Video Streaming

Airbus 09/2025 - 10/2027 Toulouse, France
R&D and Development for FTI (Flight Test Instrumentation)
Google Gemini Pro - Scrum - C++/Python - OpenGL/Unity3D
HTC Vive Pro VR virtual tour applications - Project Management

Software engineer (Real-Time Rendering and Video Codec)

HEMERIA 03/2025 - 08/2025 Angoulême, France

- Optimization of C++ code for real-time 2D overlay using ARM Mali GPU (OpenGL)
- Run with 4 separate sdi (50Hz, 1k) video channels (multi-threading)
- 17x performance gain compared to existed implementation
- FFmpeg, Cuda, nvdec, nvenc (decode and encode with overlay directly on nvidia gpus)
- IRIG
- SDI video
- Blackmagic (MultiView)
- Conan package manager

Engineer (Med Tech)

Echo3D 07/2024 - 03/2025 Toulouse, France
Real-time 3D ultrasound and AI to deliver accurate, automated diagnostic support at the point of care

- Implemented real-time 3D reconstruction from 2D ultrasound with sensor-based tracking and spatial fusion
- Built a multi-sensor calibration system, integrating : Inertial Measurement Units (IMUs) for probe motion tracking (freehand data)
- Created a mobile application for live image acquisition, reconstruction, and diagnosis (Android C++ application)
- Creation of a device to make an ultrasound scanner compatible with 3D for automatic diagnosis, patient care, and monitoring.
- 3D ultrasound visualization.
- 3D ultrasound calibration.
- Ultrasound phantom creation (silicon base).
- C++ 3D real-time application on Android.
- IMU and Optitrack sensors APIs.

Software engineer (DevOps Open Source)

Toulouse Tech Transfer 02/2024 - 06/2024 Toulouse, France

- Software development C++
- Open Source (Github, Gitlab)
- CI (Github Action, GitLab Tasks)
- C++ Unit testing (quality, catch2)
- C++ documentation (Doxygen, Sphinx)

LANGUAGES

English B2
French Native

TECHNICAL SKILLS

C (10 years)	Unix Bash (10 years)
Git (10 years)	C++ (7 years) Qt (7 years)
OpenGL ES/3/4 (7 years)	CMake (7 years)
C#/Unity3D (5 years)	Blender (5 years)
Python (5 years)	HoloLens 2/AR (2 years)
Unreal Engine 5 (2 years)	Vulkan (1 year)
Direct3D12 (1 year)	Unigine (1 year)
Arnold Renderer (1 year)	
Maya MEL Plugin (1 year)	Ada (1 year)

PUBLICATIONS

Creation of TPLAY

TTT (Toulouse Tech Transfer)
2025
<https://www.toulouse-tech-transfer.com/fr/actualites-et-evenements/t-play>

Creation of TPLAY

IRIT (Institut de Recherche en Informatique de Toulouse)
2025 <https://www.irit.fr/tplay/>

Clone of CS 1.6 with Unreal Engine

[Github](#)
2025 <https://github.com/hiergaut/CounterStrike1.6>

A Framework for Real-Time Visualization in Augmented Reality

IEEE
2024 <https://ieeexplore.ieee.org/document/10793459>

Creation of 3D fusion sensors communication protocol (TCP/IP C++ Sockets)

[Github](#)
2024 <https://github.com/T-PLAY/hub>

WORK EXPERIENCE



Research engineer (Medical Application)

IRIT (Institut de Recherche en Informatique de Toulouse)
10/2021 - 06/2024 Toulouse, France

- 3D ultrasound visualization
- 3D ultrasound calibration
- ultrasound phantom creation (gelatin base)
- 3D printing assembly (FreeCad, SolidEdge, Blender)
- FDM and resin printing
- Fast sensor flow synchronization (TCP/IP C++)
- Virtual and augmented visualization (OpenGL, Unity3D, HoloLens 2, MRTK)
- Software development (C++, CMake, Qt, OpenGL)
- Polhemus Patriot, Polhemus Viper, Intel LiDAR L515 sensors APIs



Research engineer (Offline Rendering)

IRIT (Institut de Recherche en Informatique de Toulouse)
10/2020 - 10/2021 Toulouse, France

- Shaders creation for the Arnold (API C) renderer
- Implement rendering research paper (Global Illumination Shadow Layers)
- Creation of plugin for the Maya software
- Composer user interface oriented (Natron, integrated Maya plugin)



Computer graphics programmer (Real-Time Rendering)

DIGINEXT 03/2020 - 08/2020 Toulouse, France

- Test on the possibility of integrating bgfx library allowing abstraction from the various graphics APIs (D3D11, D3D12, OpenGL, Metal, Vulkan, GNM) on an existing industrial C++ OpenGL engine
- +140% (Hz) application performance obtained
- Redesign of the Vertigo (VirtualGeo) custom rendering engine (google earth like)



Aeronautical adjuster

Airbus 01/2013 - 12/2015 Toulouse, France

- Assembly adjuster on Airbus FAL (A380 Lagardère, A41J, A50)
- Adjuster for suppliers in the Toulouse area (workshops, assembly line)
- 2x8, 3x8



Military training

Armée de Terre 01/2011 - 12/2012 Toulouse, France

- Training for french army test (Luc léger, Pull-up, Swimming)
- Commando training course in Vannes (France)
- First aid course with the Red Cross

PUBLICATIONS

Efficient Stratified 3-D Scatterer Sampling for Freehand Ultrasound Simulation

[Pubmed](#)

2023 <https://pubmed.ncbi.nlm.nih.gov/37824323/>

Creation of custom rendering engine (OpenGL, Qt, C++)

[Github](#)

2020 <https://github.com/hiergaut/Goliath-Engine>

EDUCATION



MCA (Master of Computer Applications), Computer Software Engineering, European Master's Degree in IT Expertise: Intelligent Applications and Big Data

[Gaming Campus Paris](#)

09/2025 - 09/2027



MSc Video Game Developer, Real-Time 3D Rendering and Immersive Technologies AR/VR

[Gaming Campus Paris](#)

09/2025 - 09/2027



Master's degree, IGAI (Computer Graphics and Image Processing)

[Université Paul Sabatier Toulouse III](#)

01/2018 - 12/2020



License in Computer Science (Math for IT)

[Université Paul Sabatier Toulouse III](#)

01/2015 - 12/2018



CQPM, Aeronautical assembler

[Lycée Saint-Exupéry \(Blagnac\)](#)

01/2011 - 12/2012



BAC STI, Mechanical engineering

[Lycée Victor Hugo](#)

01/2008 - 12/2011